1. Introduction

1.1. Purpose

- The purpose of this website is to develop an E-Commerce platform that enables users to buy products online. The platform aims to provide a seamless shopping experience for customers while offering robust management tools for administrators.

1.2. Scope

- The scope of this project includes the development of delivery management, advanced features like speech to text and real-time analytics.

1.3. Technologies to be used

- Java Spring Boot framework for backend development.
- MySQL database for data storage.
- Thymeleaf for server-side rendering of UI pages.
- Bootstrap for front-end styling.
- Razor pay payment gateway APIs for payment integration.

2. System Analysis

2.1. Identification of Need

- This system is needed to meet the growing need for a secure and convenient platform for online shopping and selling.

2.2. Functionalities

- **Functional Requirement**
  - ♦ User registration and Login.
  - ♦ Secure login with password encoding.
  - ♦ Admin management of brands, categories, and products.
  - ♦ Product viewing, editing, and deletion.
  - ♦ Cart functionality for customers.

- **Non-Functional Requirement**
  - ♦ Order processing and transaction management.
  - ♦ Pagination for displaying products.
  - ♦ Integration of payment gateways for secure transactions.

2.3. User Characteristics

- **Administrator:** Manages brands, categories, products, orders, view sales report.
- **Customer:** Registers, logs in, browses products, adds to cart, makes purchases.

2.4. Software Development Process

- The software development process for the e-commerce platform follows an incremental model, which is based on the Agile methodology. This model emphasizes iterative development and incremental releases of software features. The development process is divided into multiple increments or phases, each focused on delivering a specific set of functionalities.

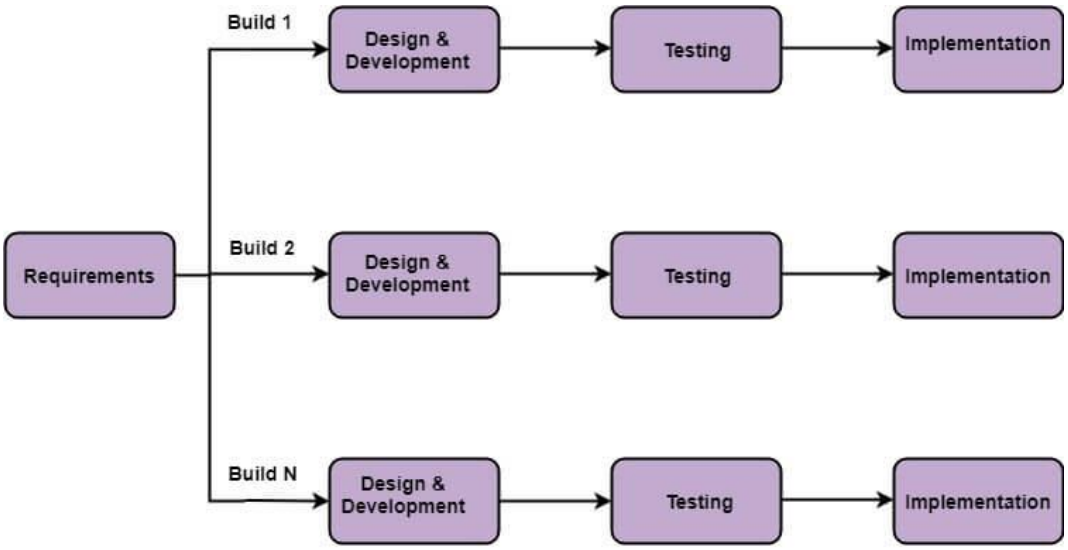


Figure 1 Incremental Model

3. Conceptual Diagram

3.1. Use Case Diagram

- **Admin use case**

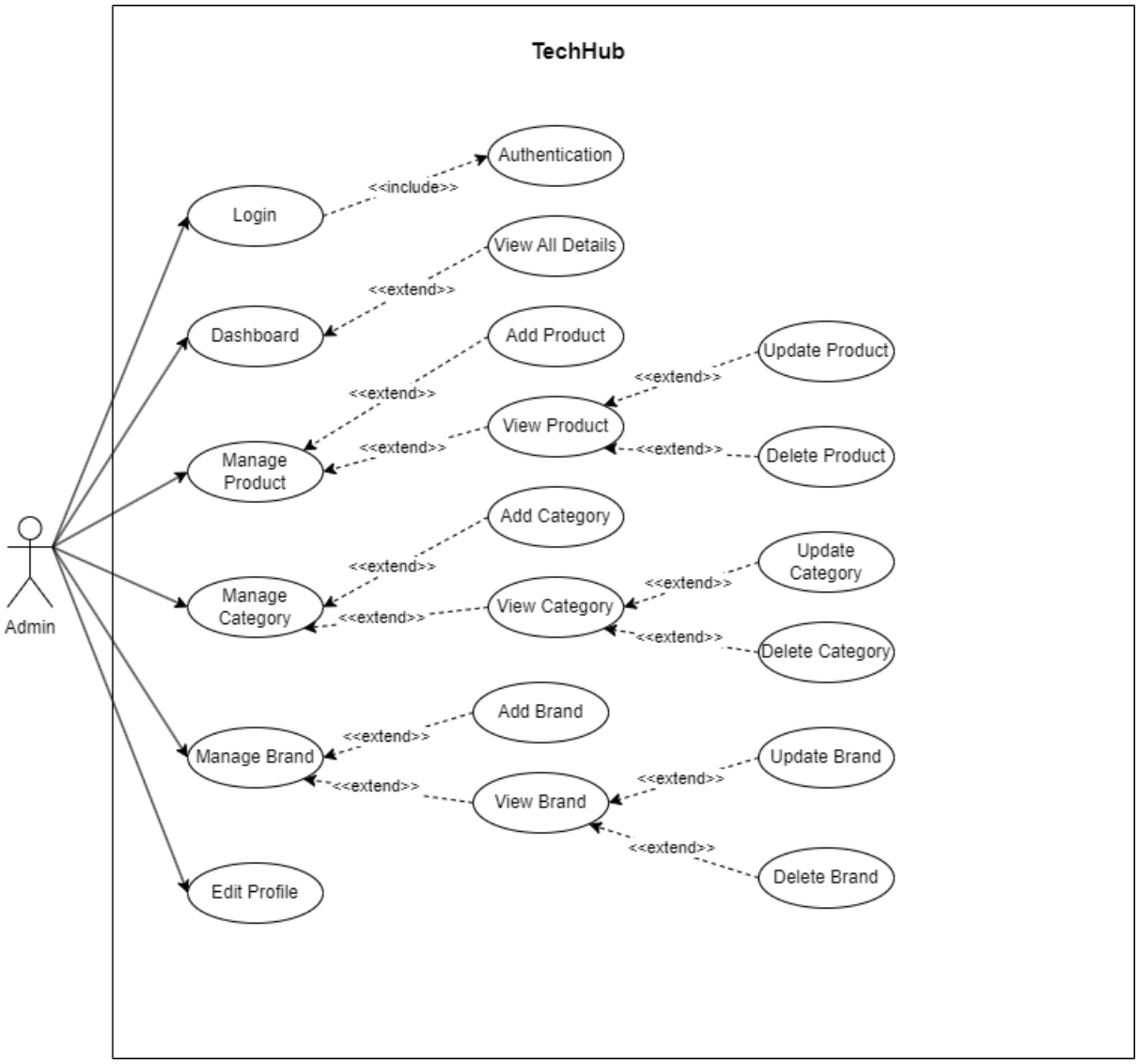


Figure 2 Admin use case diagram

- **Customer use case**

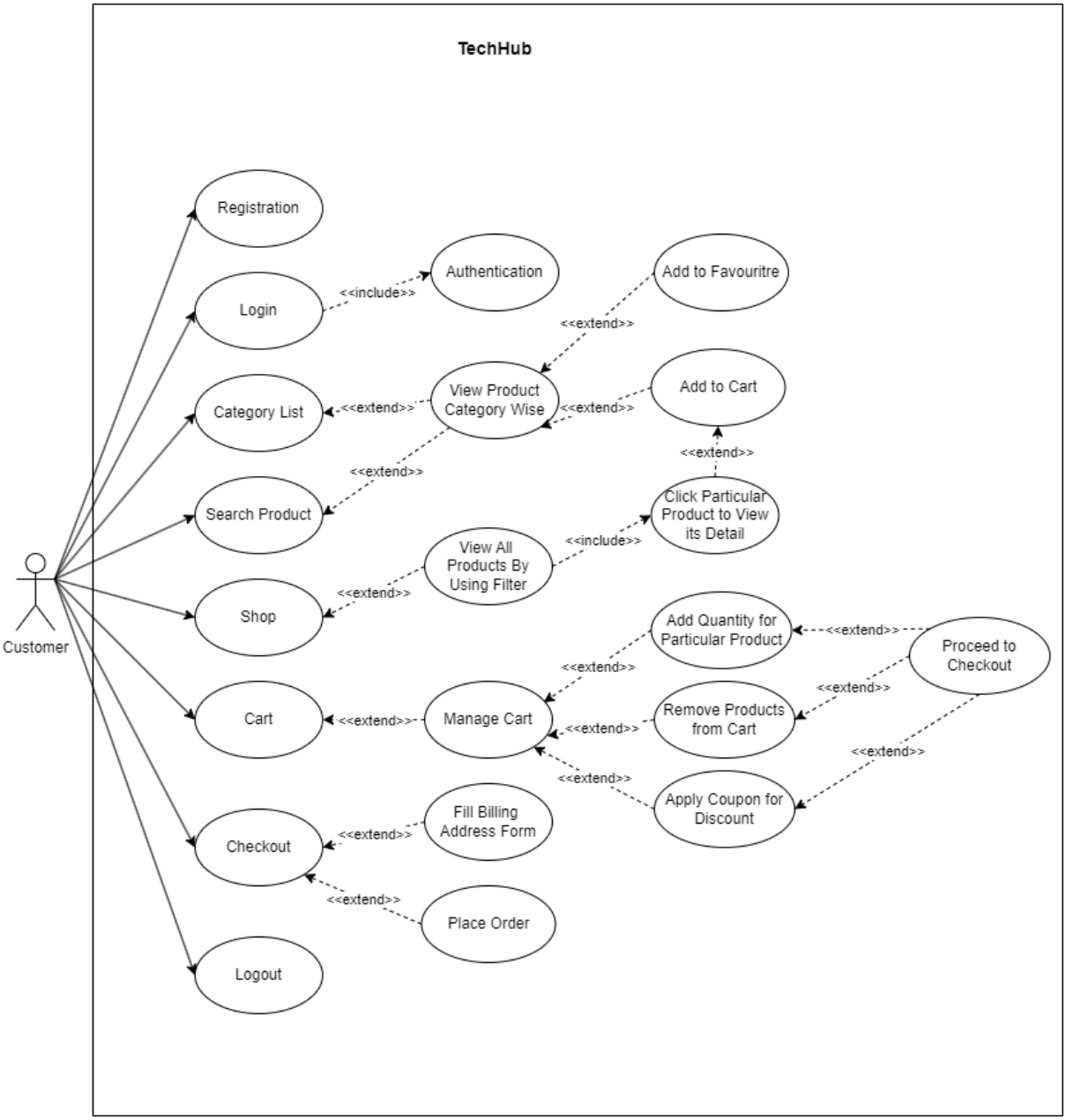


Figure 3 Customer use case diagram

3.2. Activity Diagram

- **Admin Activity Diagram**

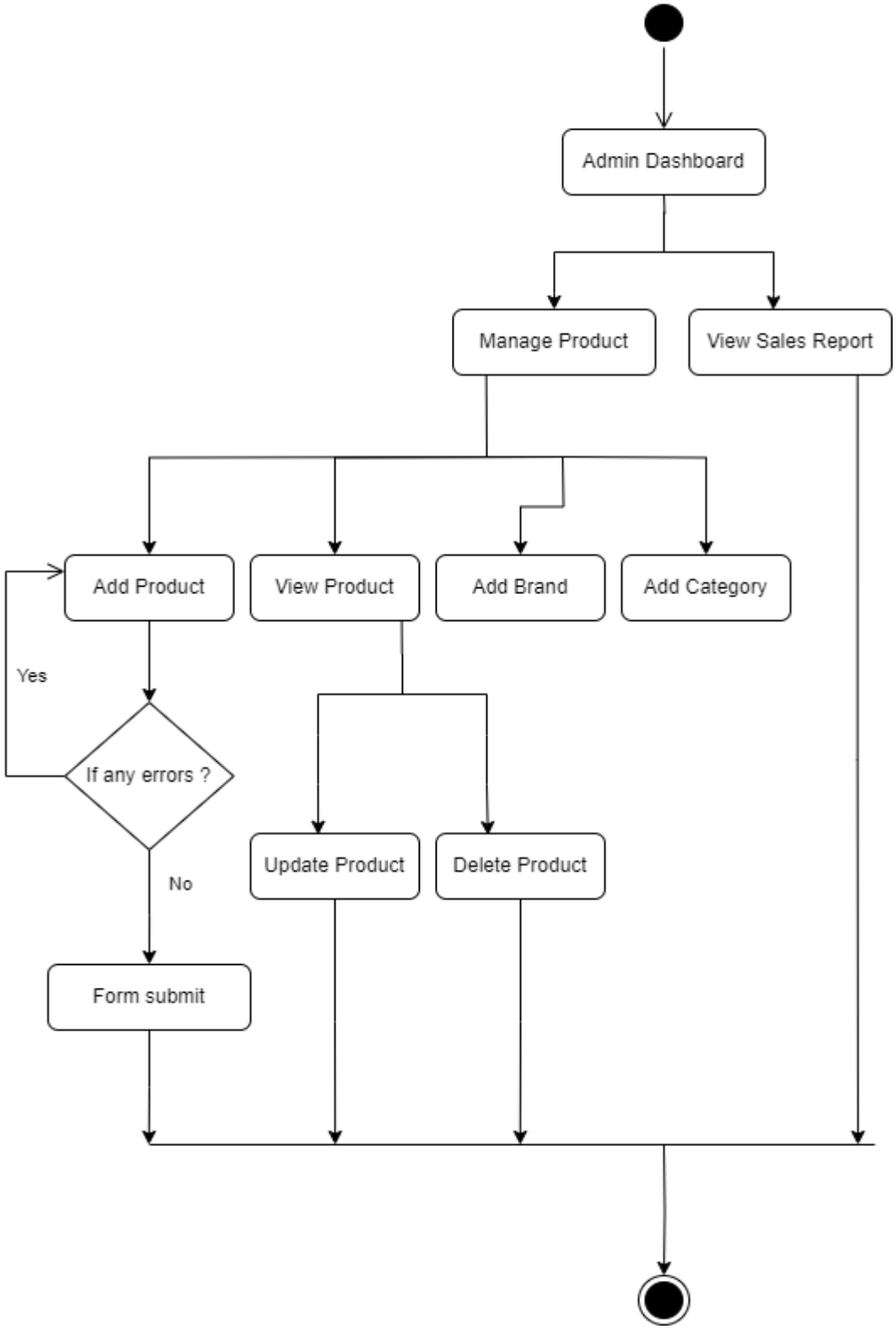


Figure 4 Admin activity diagram

- **Login Activity Diagram**

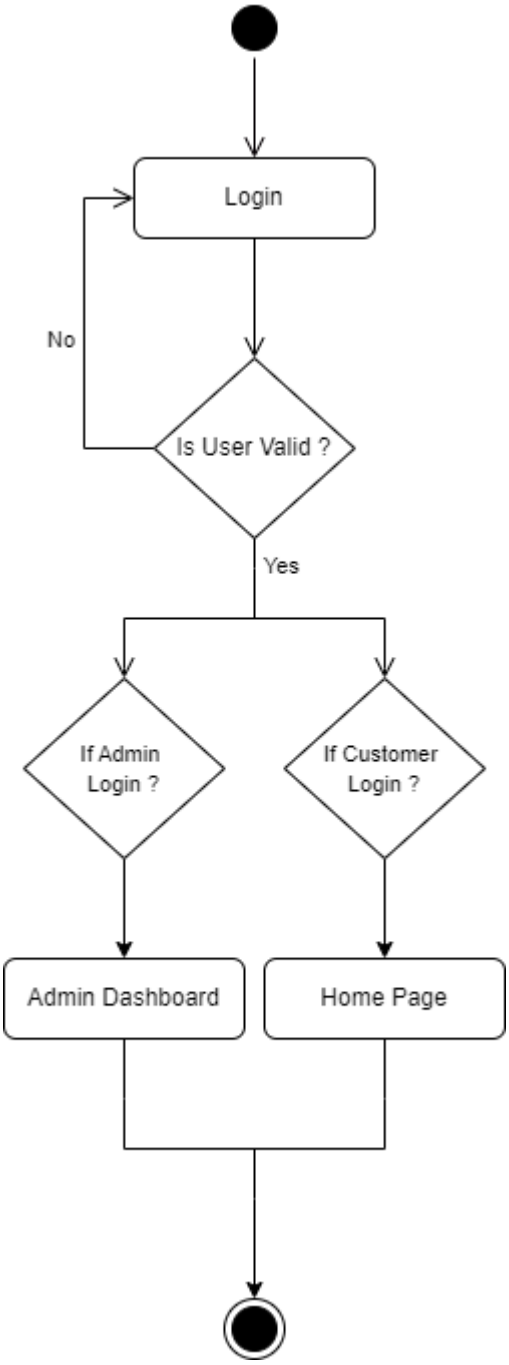


Figure 5 Login activity diagram

- **Add to Cart Activity Diagram**

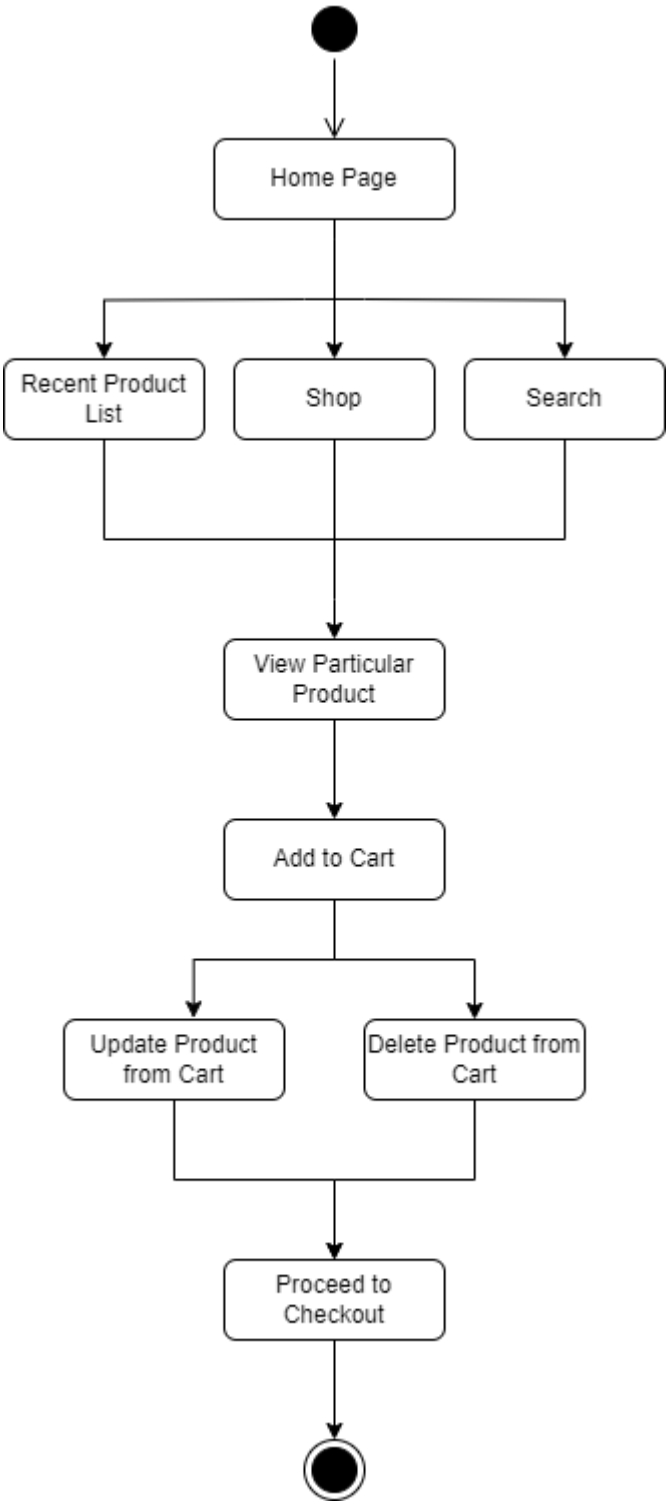


Figure 6 Add to cart activity diagram

- **Checkout Activity Diagram**

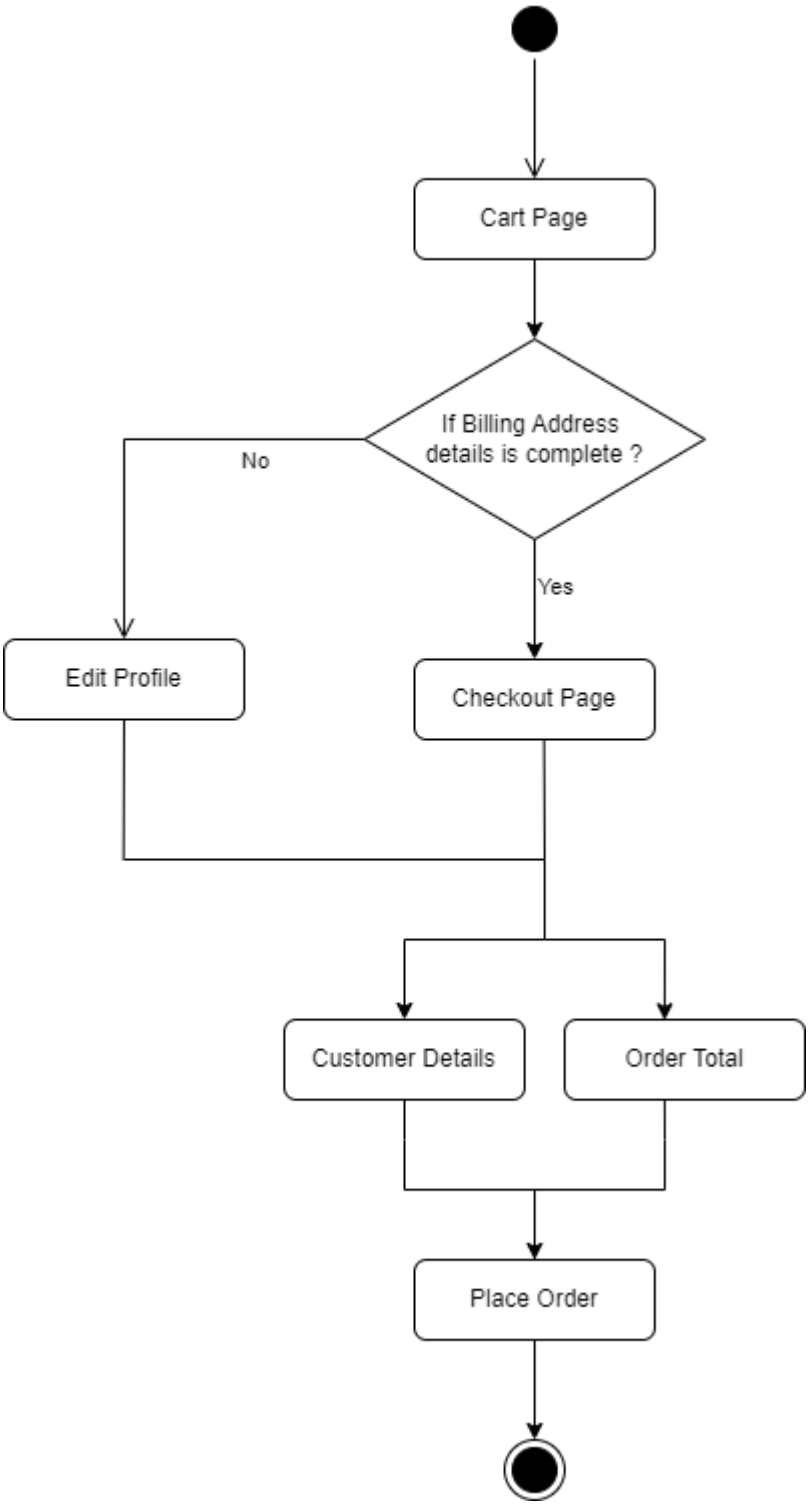


Figure 7 Checkout activity diagram

- **Payment Activity Diagram**

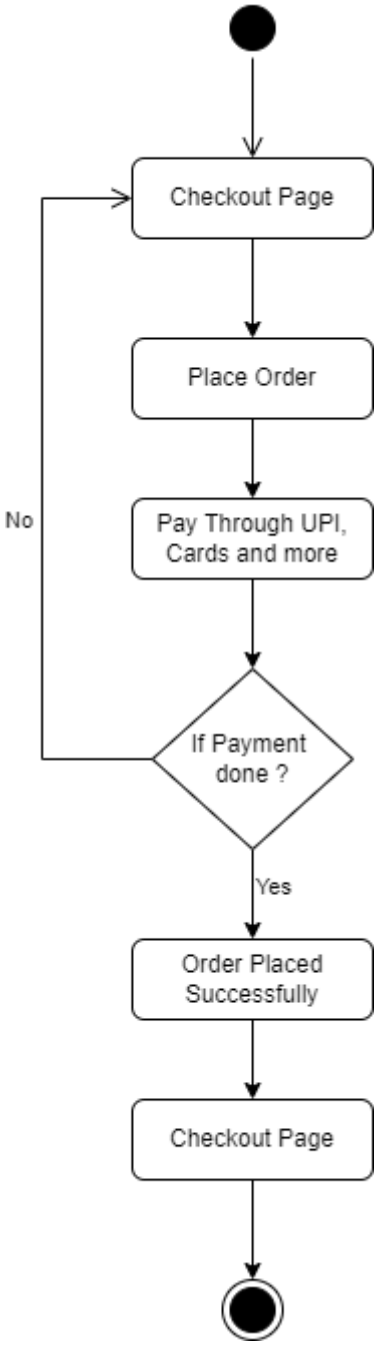


Figure 8 Payment activity diagram

4. System Design

4.1. Data Dictionary

- **Users_Master**

Field Name	Data Type	Constraints	Value	Description
User_id	bigint	Primary Key		It auto-generates the ID of the user
F_Name	Varchar (50)	Null	Uvesh	Store the user's first name
L_Name	Varchar (50)	Null	Mulla	Store the user's Last name
username	Varchar (100)	Null	uvesh07	Store the user's username
password	Varchar (255)	Null	$2a$10$xJCKWWIoKYY	Store the user's password
user_email	Varchar (255)	Null	uveshm@gmail.com	Store the user's email
phone_number	Char (10)	Null	8200988776	Store the user's phone number
user_dob	datetime (6)	Null	2002-03-01	Store the user's date of birth
user_Role	Varchar (15)	Null	Admin	User role
address	Varchar (255)	Null	MG Road	Store the user's address
city	Varchar (255)	Null	Valsad	Store the user's city
state	Varchar (255)	Null	Gujarat	Store the user's state
country	Varchar (255)	Null	India	Store the user's country
ZipCode	Varchar (255)	Null	396001	Store the user's zip code

- **Brand_Master**

Field Name	Data Type	Constraints	Value	Description
brand_id	bigint	Primary Key		It auto-generates the ID of the brand
brand_name	Varchar (255)	Not Null	Dell	Store the brand name
Country_of_Origin	Varchar (100)	Null	China	Store the country name

- **Category_Master**

Field Name	Data Type	Constraints	Value	Description
category_id	bigint	Primary Key		It auto-generates the ID of the category
categoryname	Varchar (100)	Not Null	Laptop	Store the category name

- **Cart_Master**

Field Name	Data Type	Constraints	Value	Description
cart_id	bigint	Primary Key		It auto-generates the ID of the cart
total_items	int	Not Null	1	Store the total item in cart
total_prices	double	Not Null	49999	Store the total price of cart
user_id	bigint	Unique Key	2	Store the user's user_id

- **Cart_Item**

Field Name	Data Type	Constraints	Value	Description
cart_item_id	bigint	Primary Key		It auto-generates the ID of the cart item
qunatity	int	Not Null	1	Store the quantity
total_price	double	Not Null	49999	Store the total price
unit_price	double	Not Null	49999	Store the unit price
cart_id	bigint	Foreign Key	1	Store the cart_id
product_id	bigint	Unique Key	15	Store the product_id

- **Product_Master**

Field Name	Data Type	Constraints	Value	Description
product_id	bigint	Primary Key		It auto-generates the ID of the product
product_name	Varchar (255)	Not Null	Dell Inspiron 13	Store the product name
category_id	Varchar (50)	Foreign Key	1	Store the category_id
brand_id	Varchar (100)	Foreign Key	1	Store the brand_id
product_image	Varchar (255)	Not Null	1709104070598_Dell13.jpg	Store the image path
product_price	double	Not Null	49999	Store the product price

Manufacturer	Varchar (255)	Not Null	Dell	Store the manufacturer name
Series	Varchar (100)	Not Null	D134JU-N3201WS	Store the product series
colour	Varchar (100)	Not Null	Gray	Store the product colour
Form_Factor	Varchar (100)	Not Null	laptop	Store the product form
product_Height	Varchar (100)	Not Null	2.26 CM	Store the product height
product_Weight	Varchar (100)	Not Null	2.50 Kg	Store the product weight
stockquantity	Varchar (100)	Not Null	20	Store the product stock quantity
Product_Dimensions	Varchar (100)	Null	42.1 x 10.5 x 42.1 cm 2.5 Kilograms	Store the product dimension
Standing_screen_display_size	Varchar (100)	Null	40.64 Centimeters	Store the product screen size
Screen_Resolution	Varchar (100)	Null	1920 x 1080 pixel	Store the product screen resolution
Batteries	Varchar (100)	Null	1 Lithium Ion batteries required	Store the product batteries info
Processor_Brand	Varchar (100)	Null	Intel	Store the product processor brand
Processor_Type	Varchar (100)	Null	Core i5	Store the product processor type

Processor_Speed	Varchar (100)	Null	2.4 GHz	Store the product processor speed
Processor_Count	Varchar (100)	Null	4	Store the product processor count
RAM_Size	Varchar (100)	Null	16 GB	Store the product RAM Size
Memory_Technology	Varchar (100)	Null	DDR5	Store the product memory technology
Maximum_Memory_Supported	Varchar (100)	Null	32 GB	Store the product maximum memory supported
Memory_Clock_Speed	Varchar (100)	Null	Null	Store the product memory clock speed
Hard_Drive_Size	Varchar (100)	Null	Null	Store the product hard drive size
Hard_Disk_Type_Description	Varchar (100)	Null	SSD	Store the product hard disk type
Audio_Details	Varchar (100)	Null	Headphones, Speakers	Store the product audio details
Graphics_Chipset_Brand	Varchar (100)	Null	Intel Iris	Store the product graphics chipset brand
Graphics_Card_Description	Varchar (100)	Null	Null	Store the product graphics card description
Graphics_RAM_Type	Varchar (100)	Null	Null	Store the product graphics RAM type

Graphics_Card_Ram_Size	Varchar (100)	Null	1 GB	Store the product graphics RAM size
Graphics_Card_Interface	Varchar (100)	Null	Integrated	Store the product graphics card interface
Connectivity_Type	Varchar (100)	Null	Bluetooth, Wi-Fi	Store the product connectivity type
Number_of_USB_2_0_Ports	Varchar (3)	Null	2	Store the product USB 2.0 ports
Number_of_USB_3_0_Ports	Varchar (3)	Null	1	Store the product USB 3.0 ports
Optical_Drive_Type	Varchar (30)	Null	No Optical Storage	Store the product optical drive type
Rear_Webcam_Resolution	Varchar (100)	Null	2560x1600 pixels	Store the product front webcam resolution
Power_Source	Varchar (100)	Null	Battery Powered	Store the product power source
Operating_System	Varchar (100)	Null	Windows 11 Home	Store the product O.S
Number_of_Lithium_Ion_Cells	Varchar (3)	Null	4	Store the product lithium ion cells
Material	Varchar (255)	Null	Null	Store the product material
Movement_Detection_Technology	Varchar (100)	Null	Null	Store the product movement detection technology

Connector_Type	Varchar (100)	Null	Null	Store the product connector type
batteries_included	Varchar (255)	Null	Null	Store the product batteries included
batteries_required	Varchar (255)	Null	Null	Store the product batteries required
Hand_Orientation	Varchar (100)	Null	Null	Store the product hand orientation
Maximum_resolution	Varchar (100)	Null	Null	Store the product max resolution
Hardware_Platform	Varchar (255)	Null	Null	Store the product hardware platform
Special_feature	Varchar (255)	Null	Null	Store the product special feature
Compatible_device	Varchar (255)	Null	Null	Store the product compatible device type
RAM_Type	Varchar (100)	Null	Null	Store the product RAM type
Included_Components	Varchar (255)	Not Null	Laptop, Adapter & User Manual	Store the product included components
Country_of_Origin	Varchar (100)	Null	China	Store the product country of origin
Product_quantity	int	Not Null	55	Store the product quantity
Product_Decscription	Varchar (255)	Not Null	Great for the business work and entertainment	Store the product description

Product_Warranty	Varchar (20)	Null	2 Years	Store the product warranty info
created_at	datetime	Not Null	2024-02-28 12:37:50	Store the product creation/ added date

- **Order_Master**

Field Name	Data Type	Constraints	Value	Description
order_id	bigint	Primary Key		It auto-generates the ID of the order_id
user_id	bigint	Foreign Key	2	Store the user_id
razorOrderid	Varchar (255)	Not Null		Store the razor pay order id
razorPyamentid	Varchar (255)	Not Null		Store the razor pay payment id
order_date	Datetime	Not Null	2024-03-28 13:37:50	Store the order date
Total_Amount	double	Not Null	49999	Store the total amount
currency	Varchar (255)	Not Null	INR	Store the currency
status	Varchar (255)	Not Null		Store the status
receipt	Varchar (255)	Not Null		Store the receipt

4.1. User Interface Design

- **Registration Page**

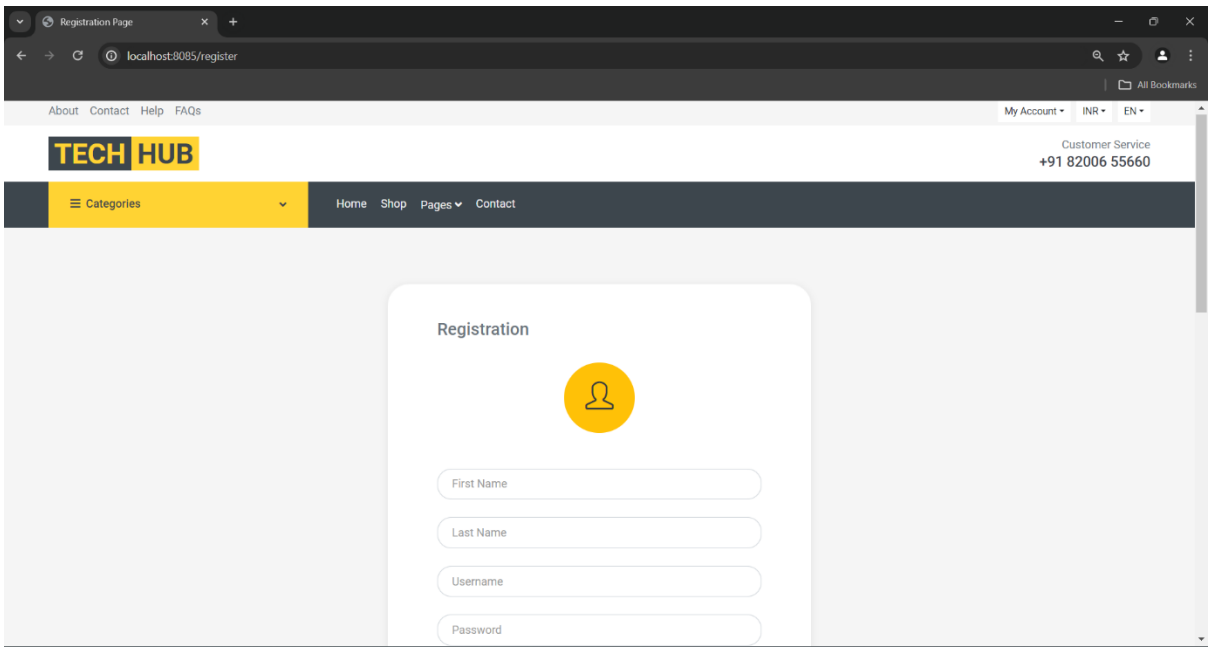


Figure 9 Registration page

- **Login Page**

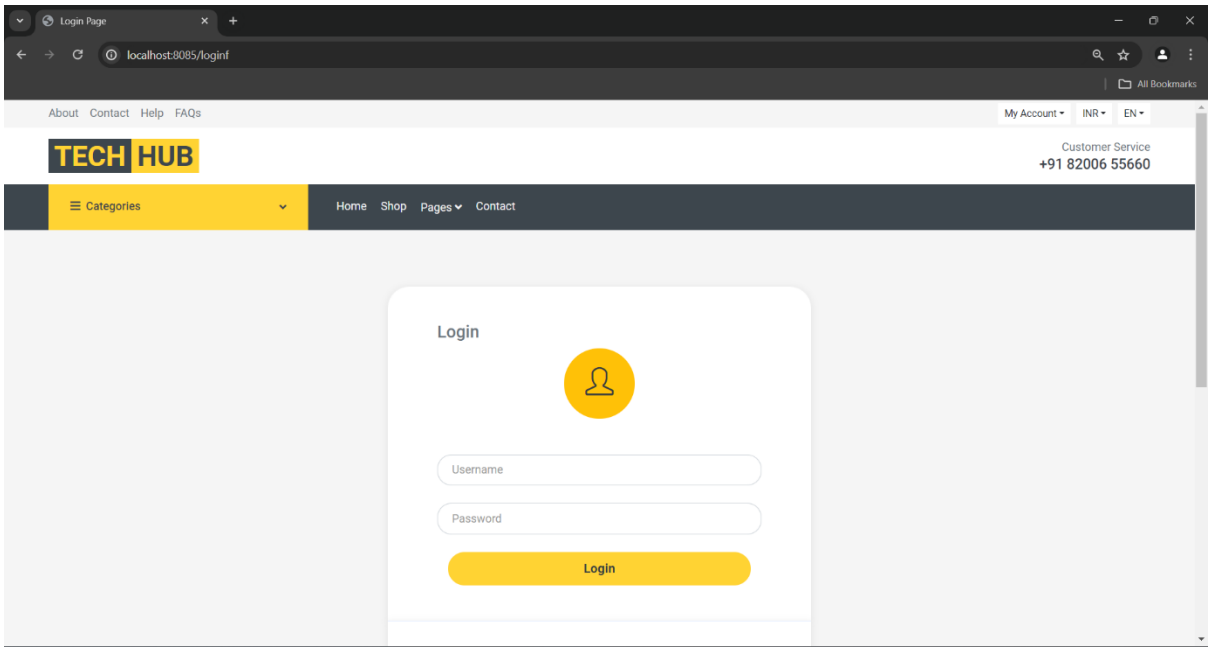


Figure 10 Login page

- **Home Page**

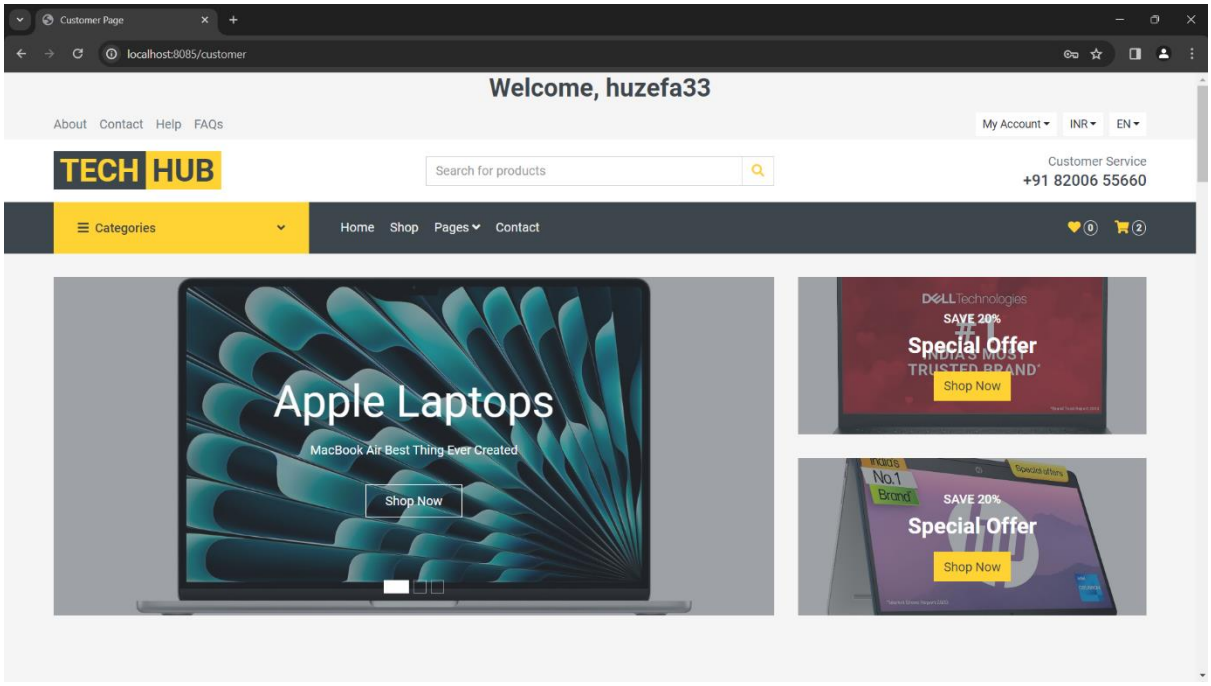


Figure 11-1 Home page

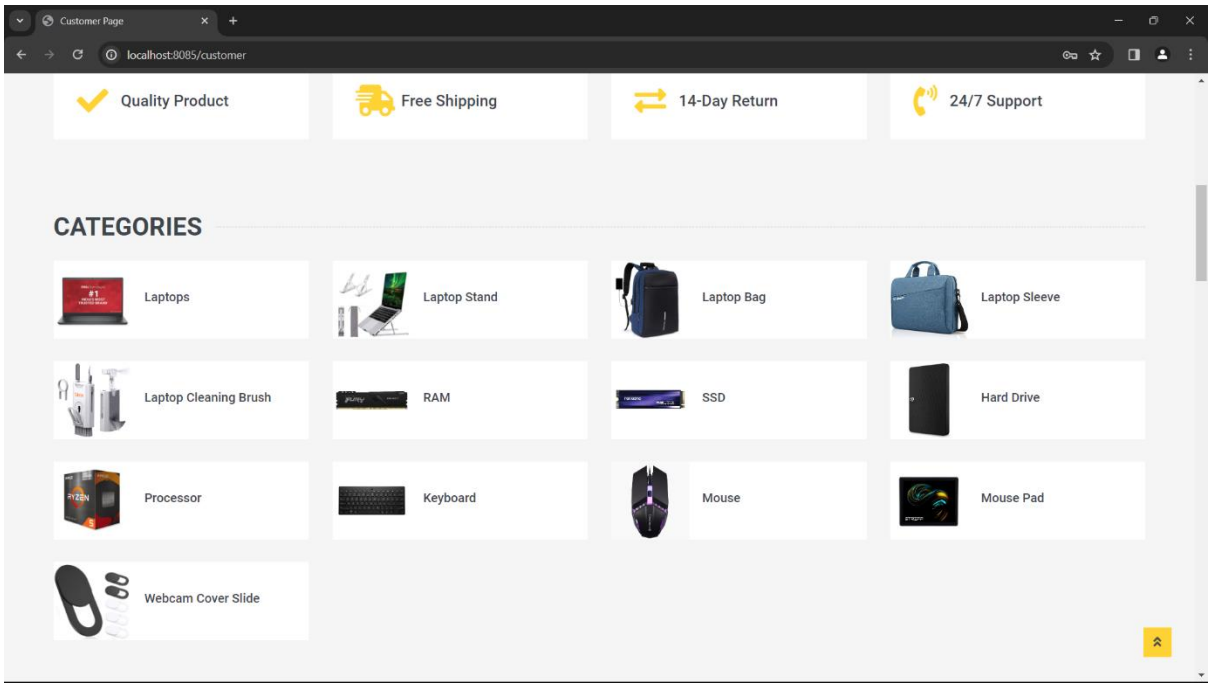


Figure 12-2 Home page

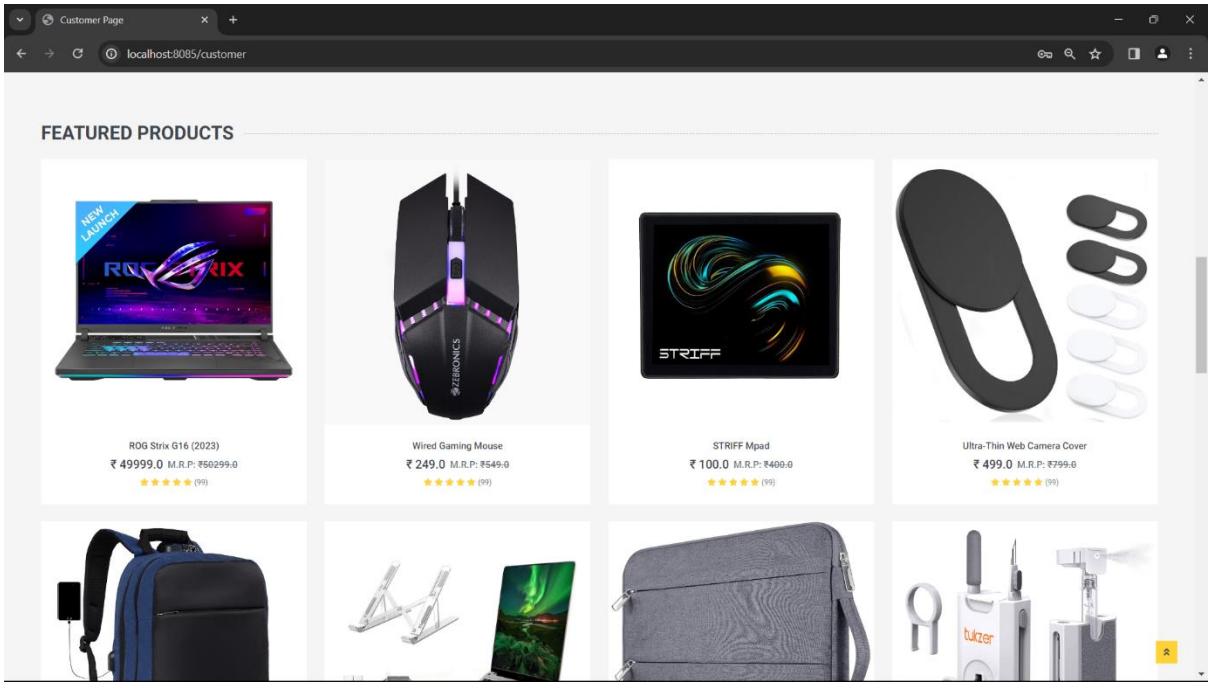


Figure 13-3 Home page

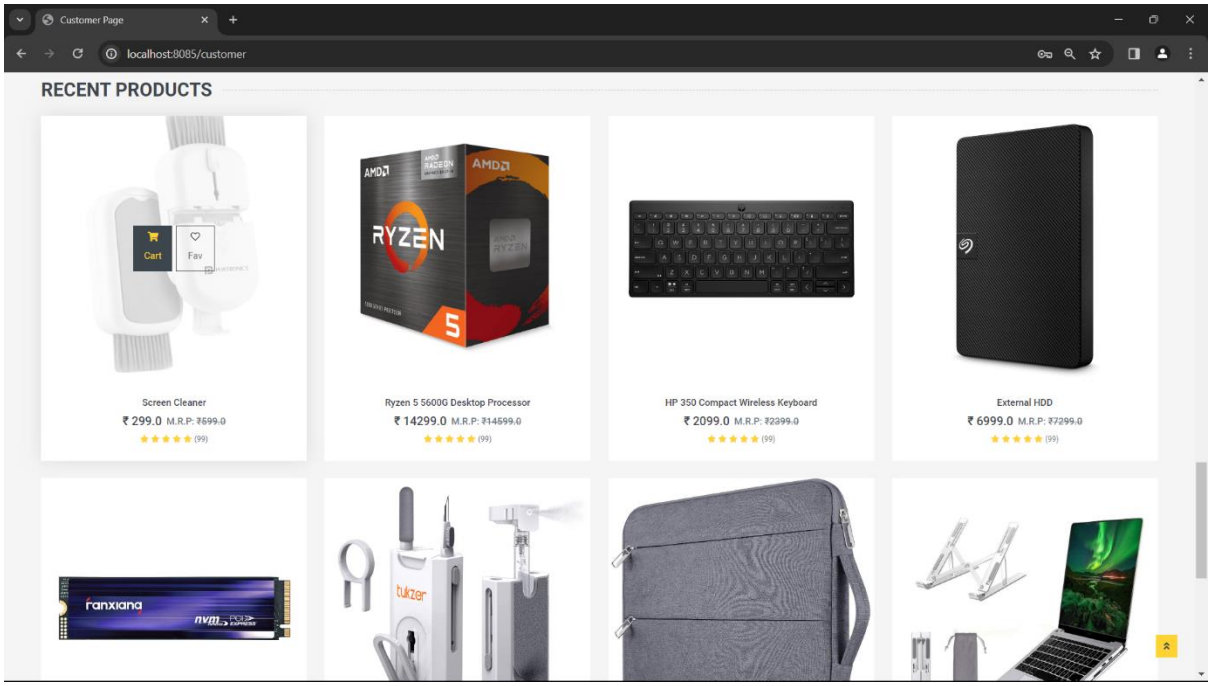


Figure 14-4 Home page

- **Shop Page**

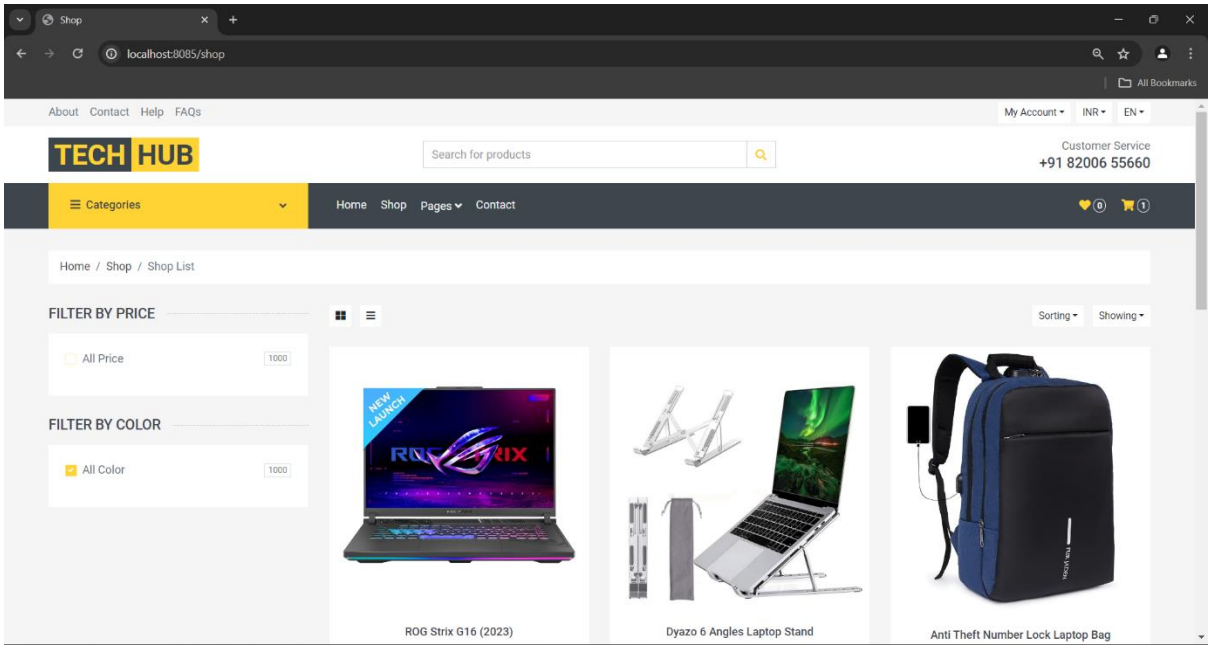


Figure 15-1 Shop page

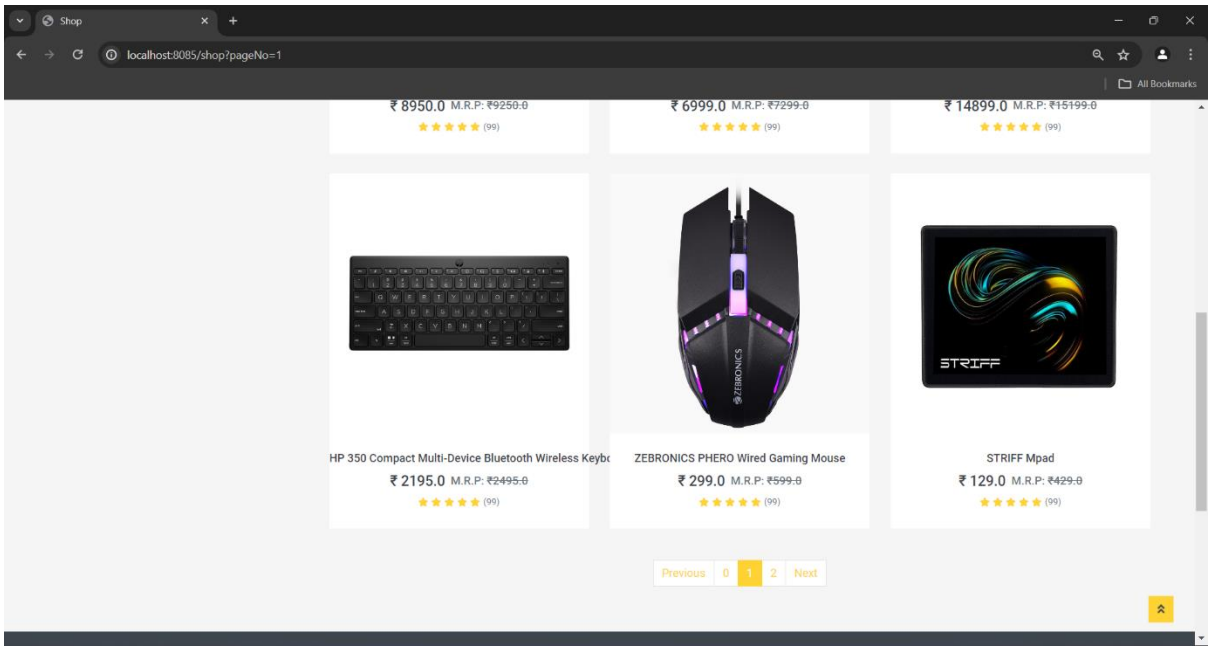


Figure 16-2 Shop page

- **Cart Page**

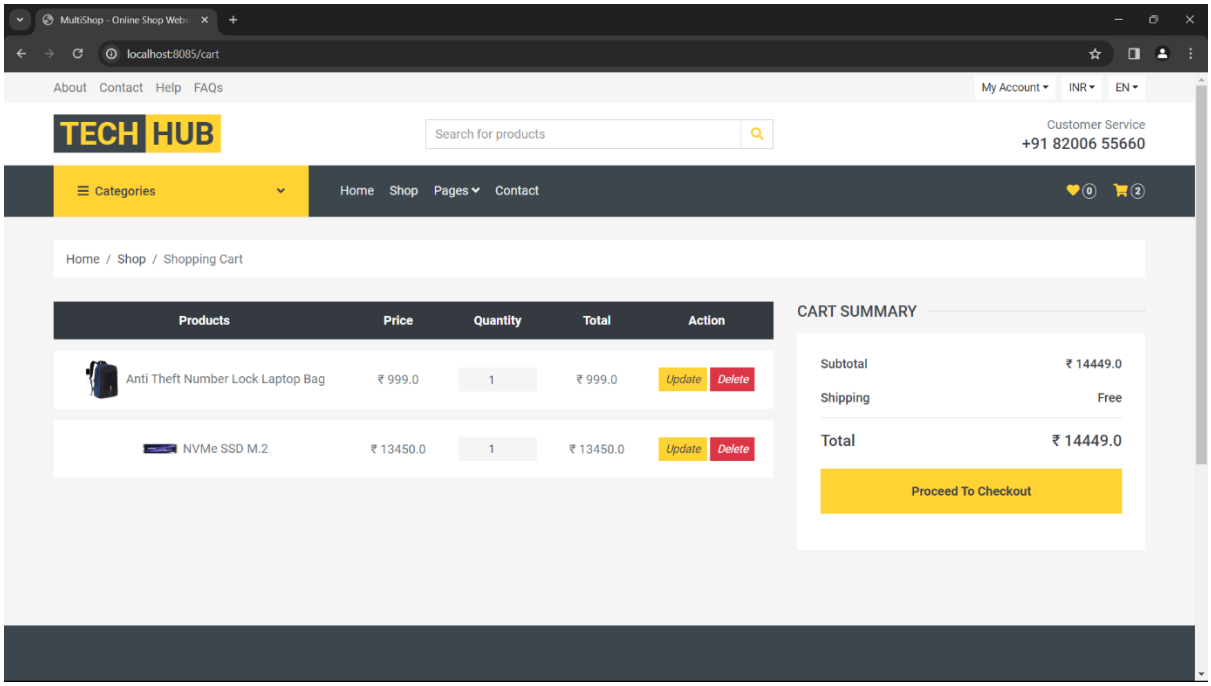


Figure 17 Cart page

- **Shop/Product Detail Page**

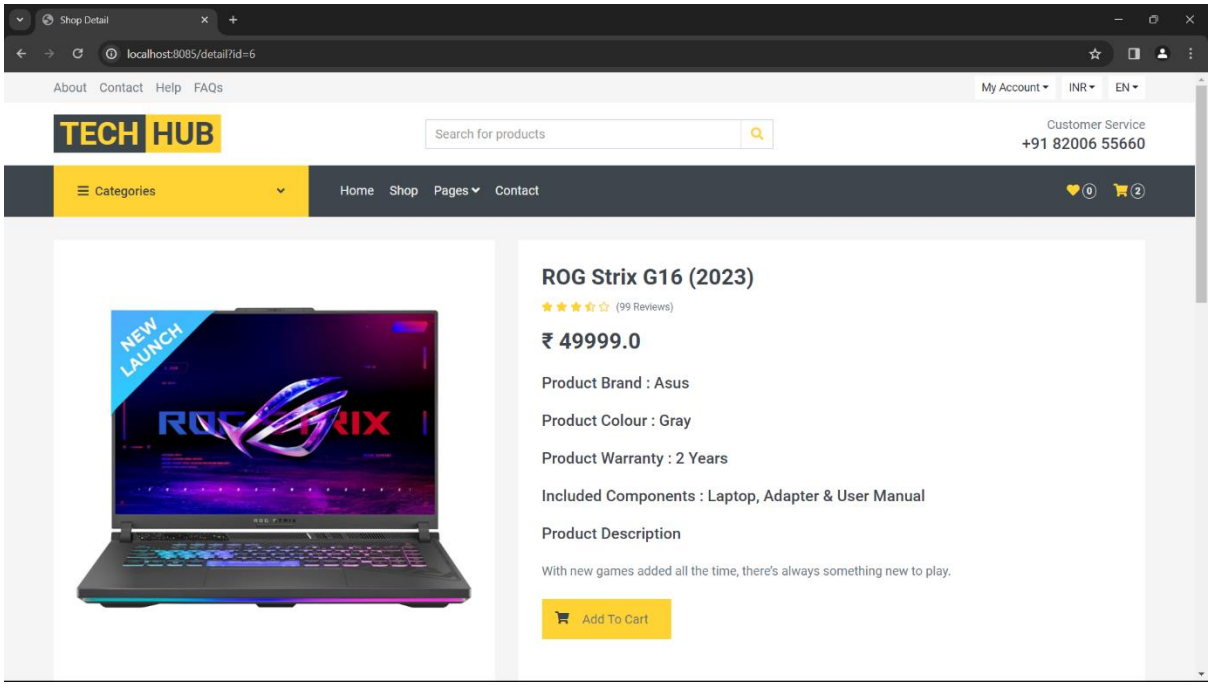


Figure 18-1 Shop/product detail page

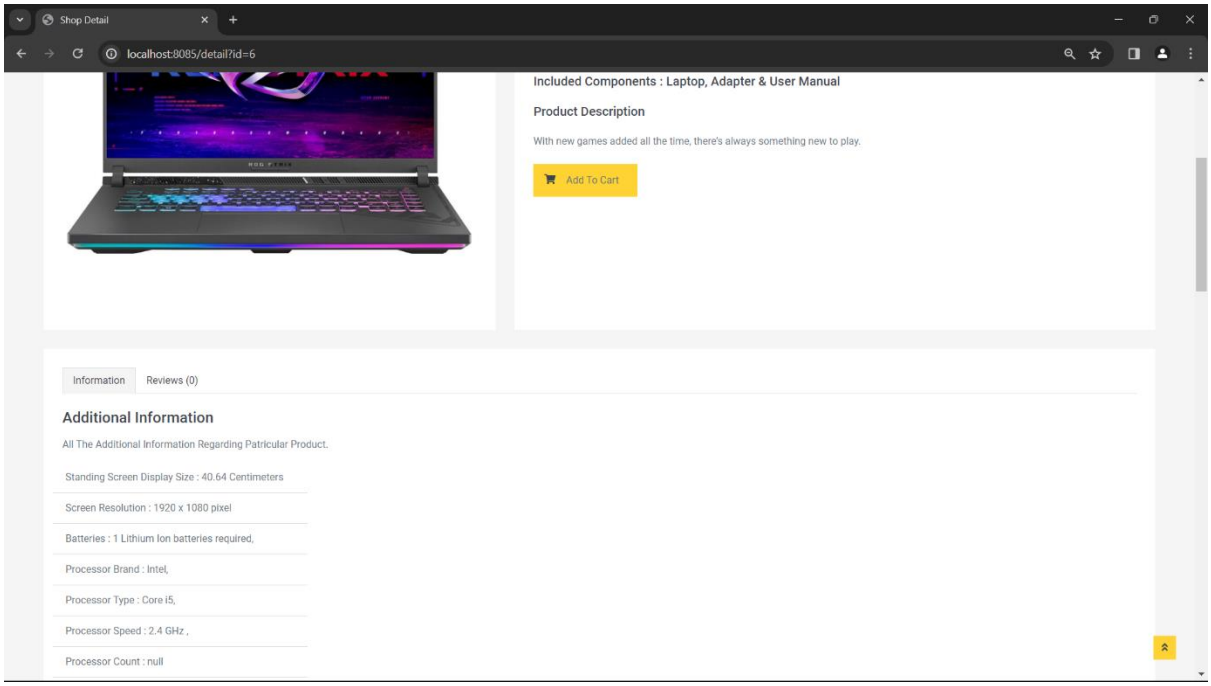


Figure 19-2 Shop/product detail page

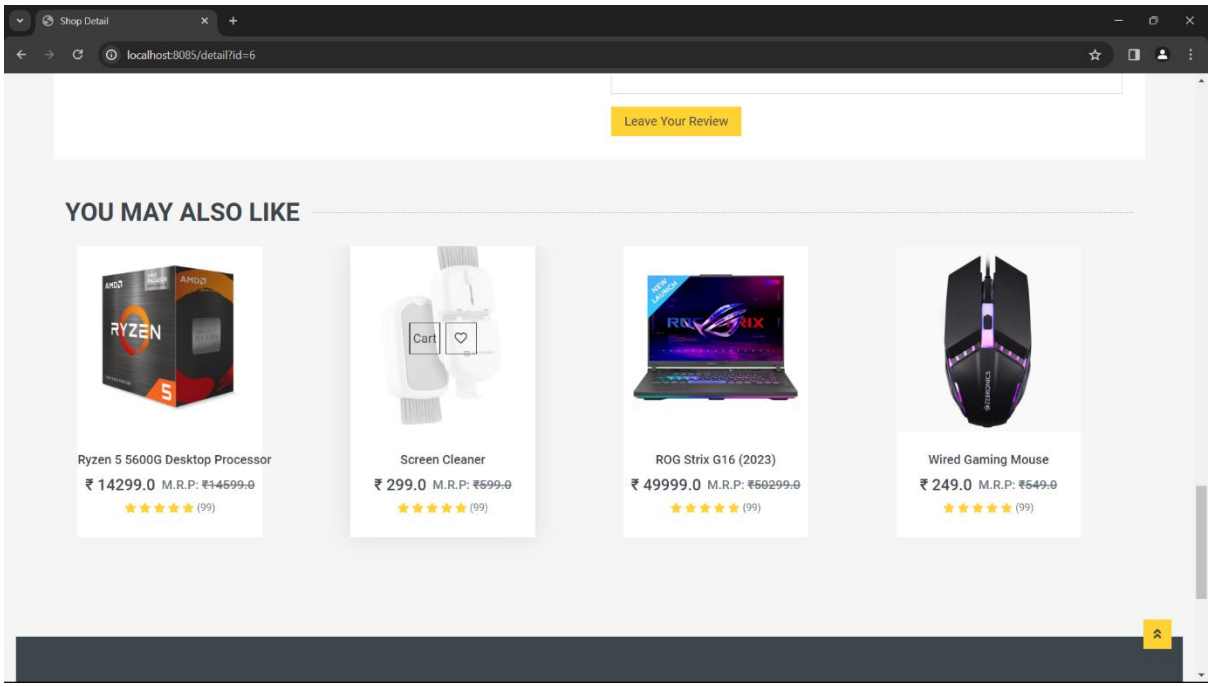


Figure 20-3 Shop/product detail page

- **Checkout Page**

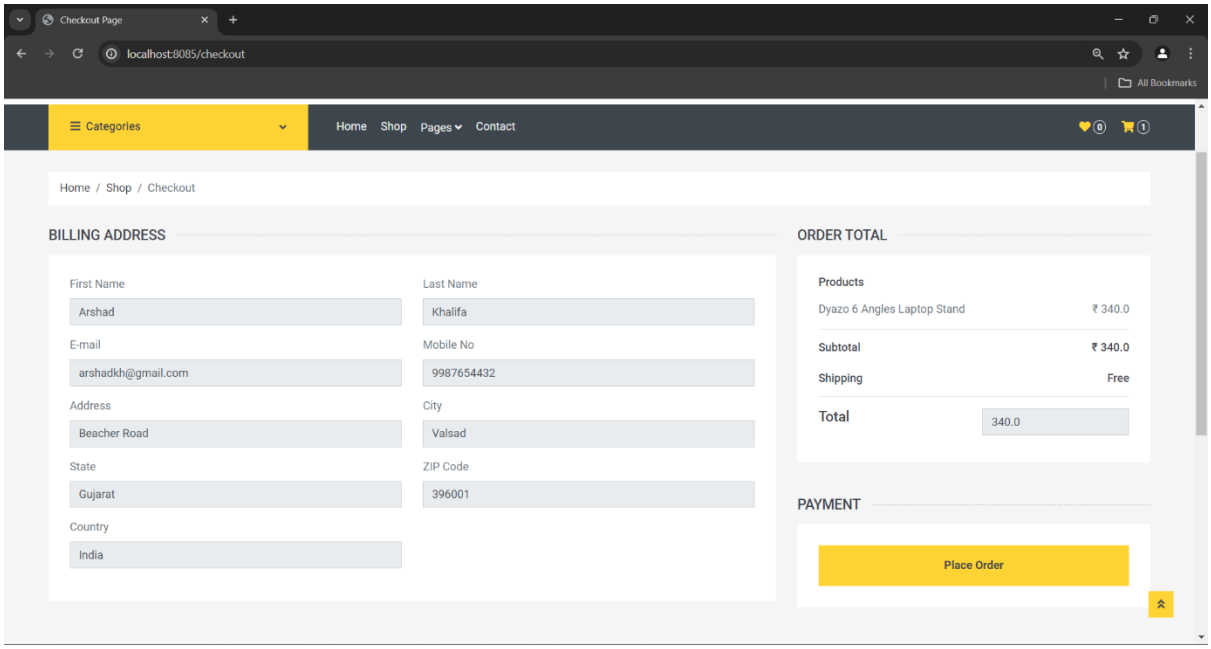


Figure 21 Checkout page

- **Place Order**

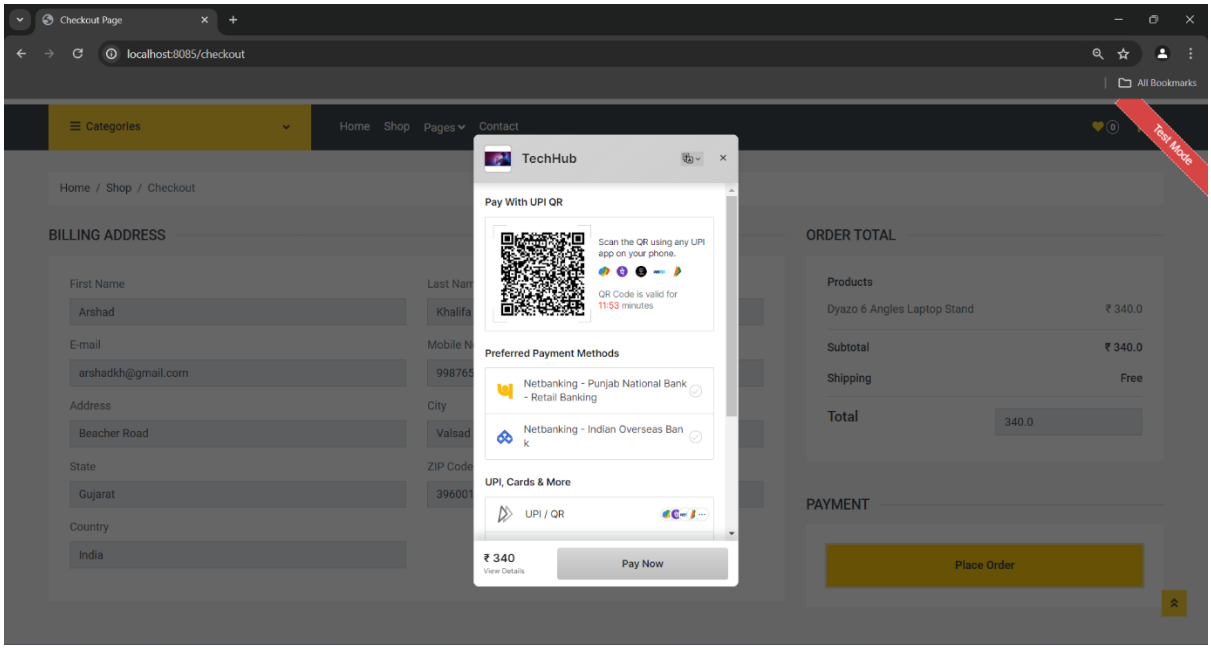


Figure 22-1 Place order

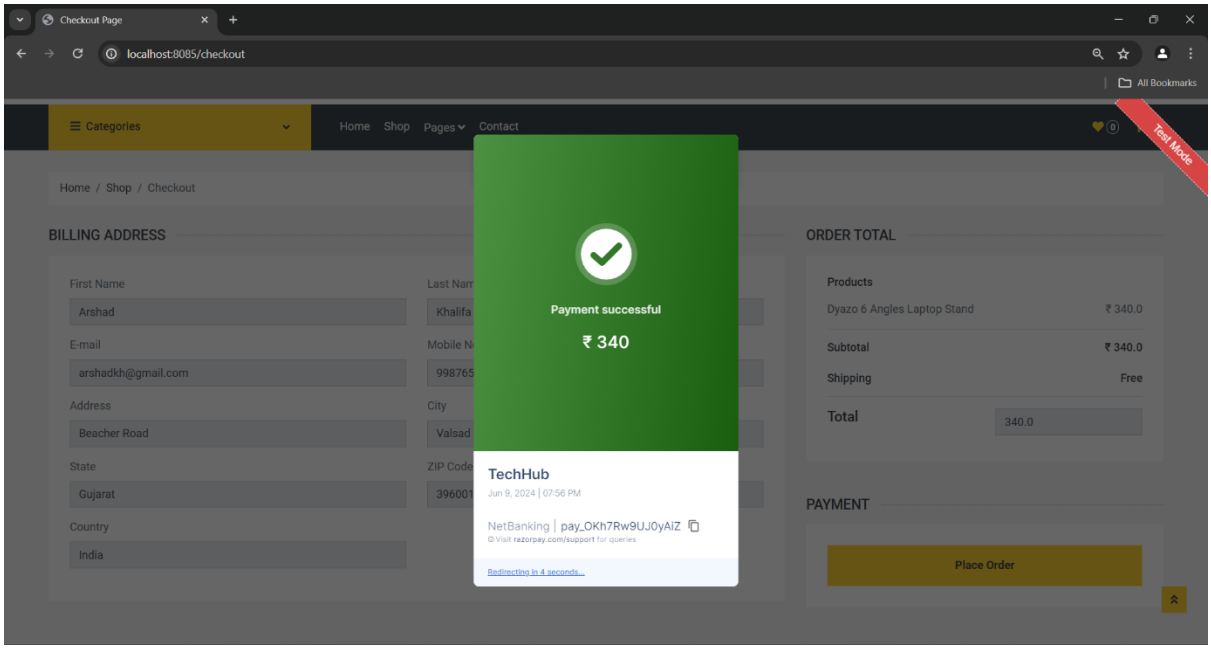


Figure 23-2 Place order

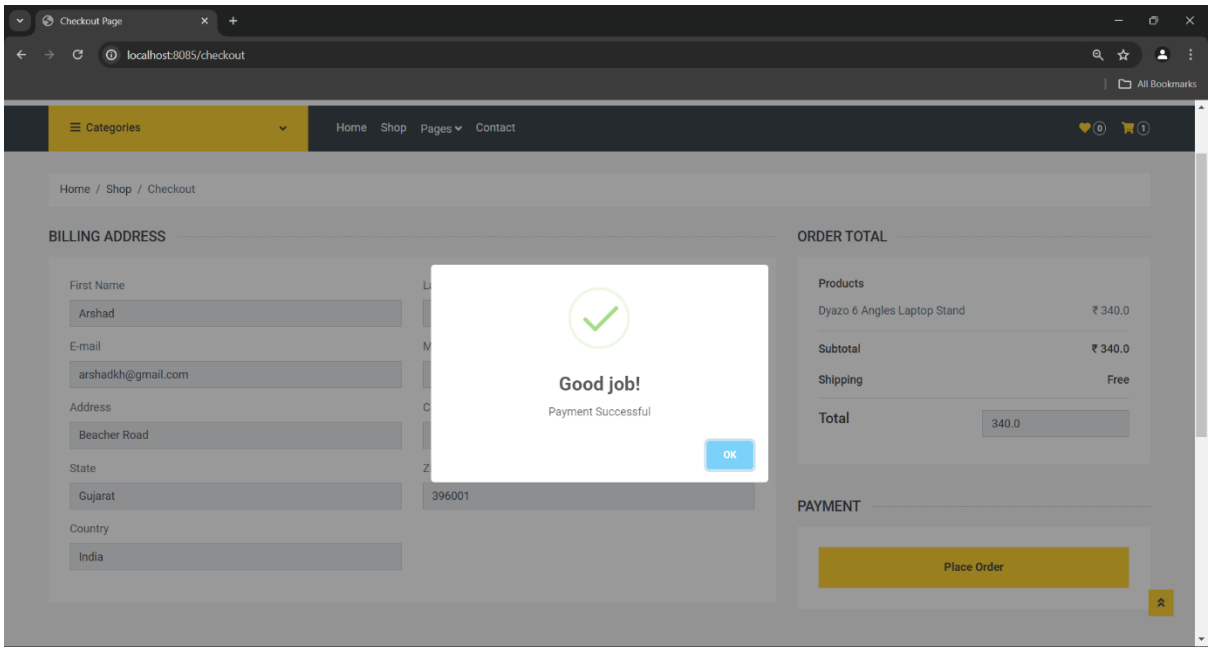


Figure 24-3 Place order

4.2. Admin Dashboards/ Reports

- **Add Brand Page**

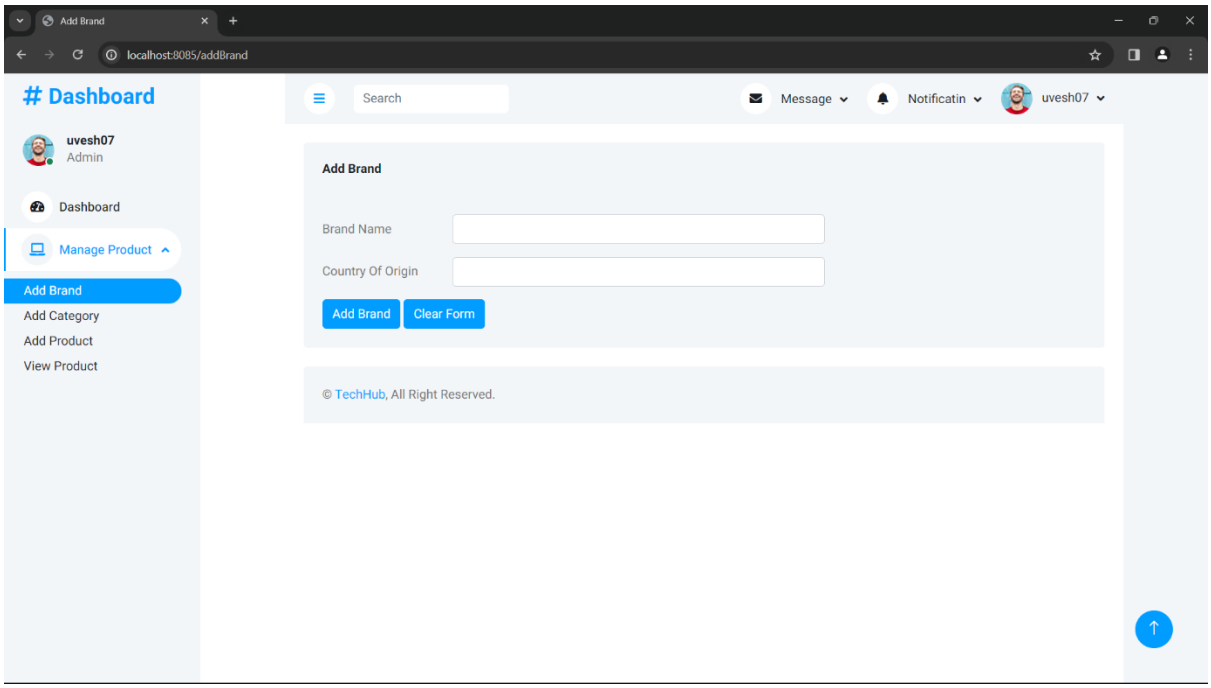


Figure 25 Add brand page

- **Add Category Page**

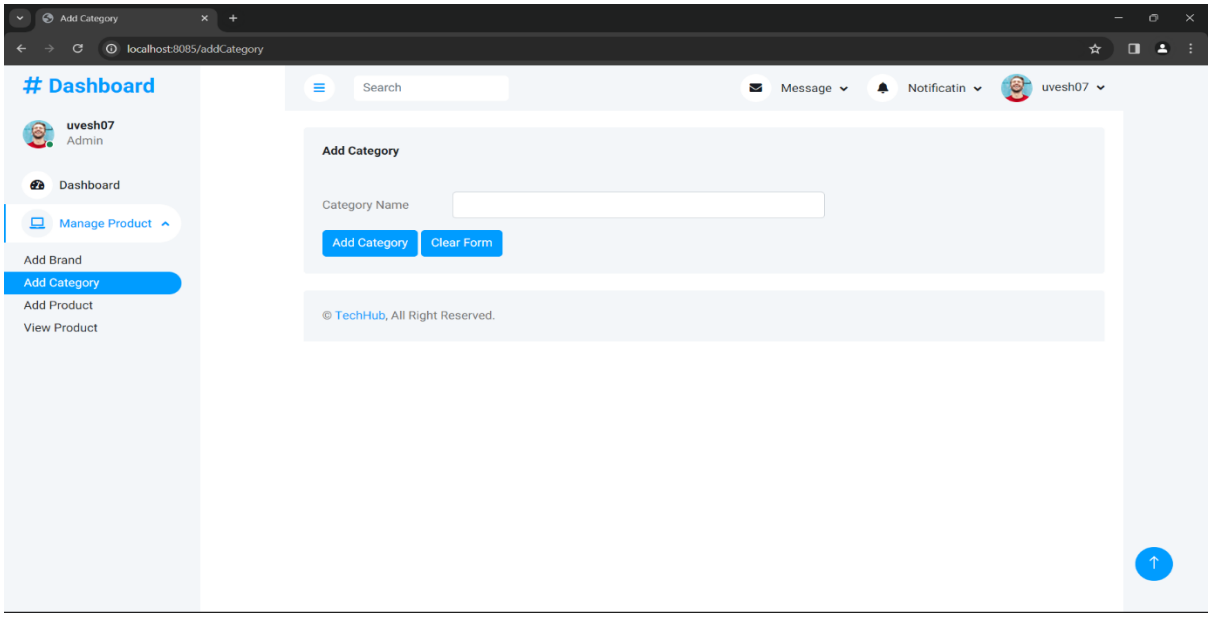


Figure 26 Add category page

- **Add Product Page**

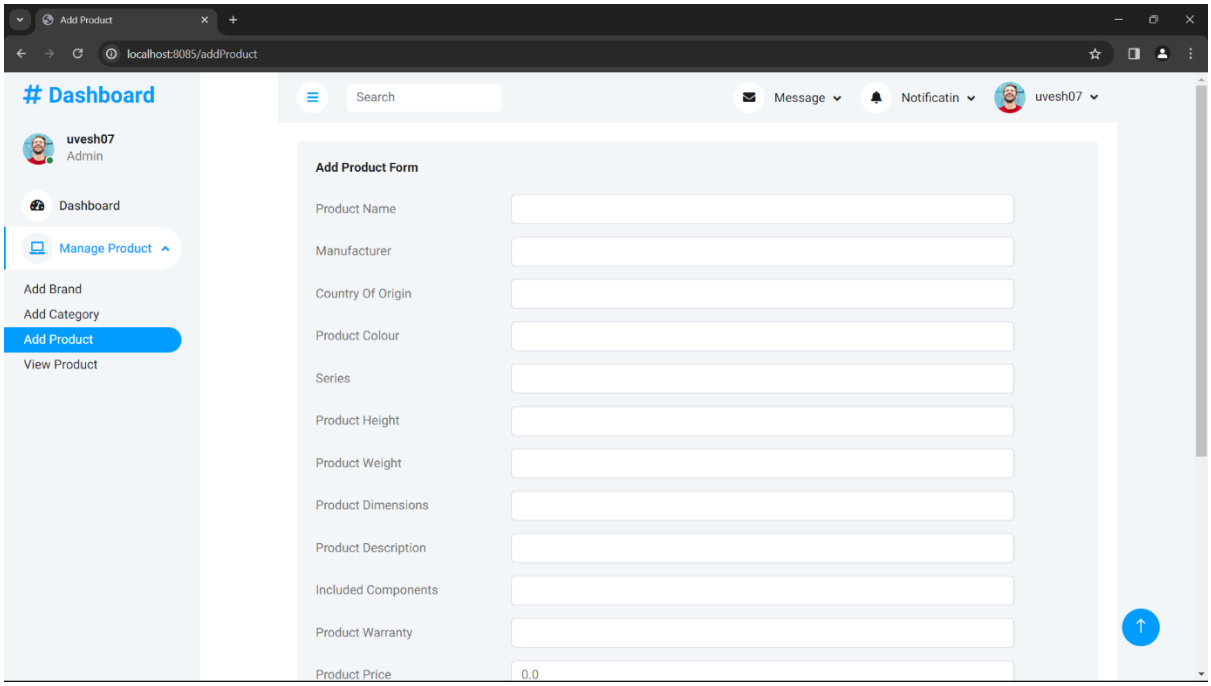


Figure 27 Add product page

- **View Product Page**

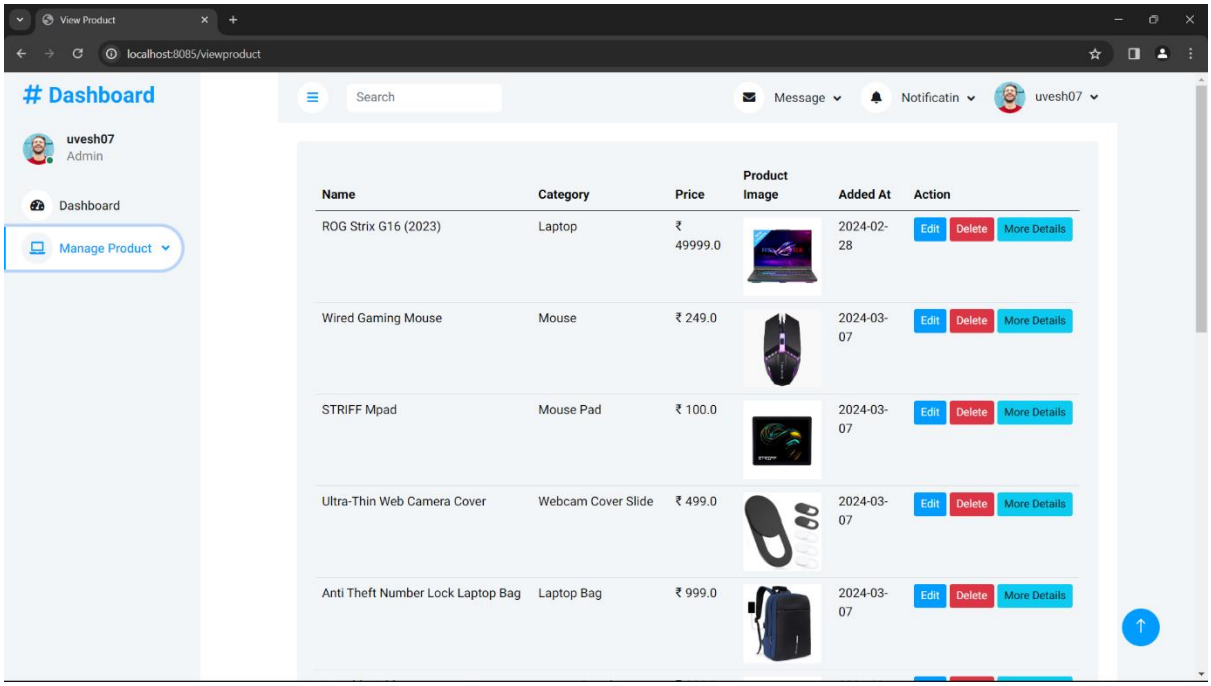


Figure 28 View product page

- **Edit Product Page**

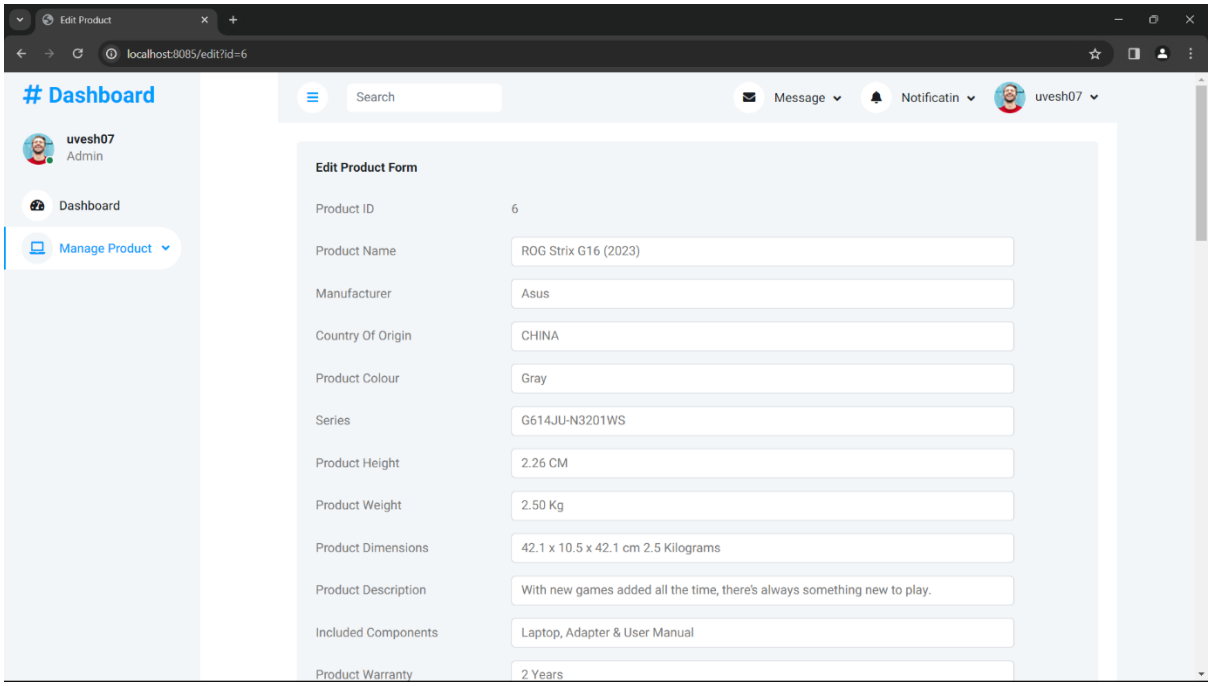


Figure 29 Edit product page

- **More Details Page**

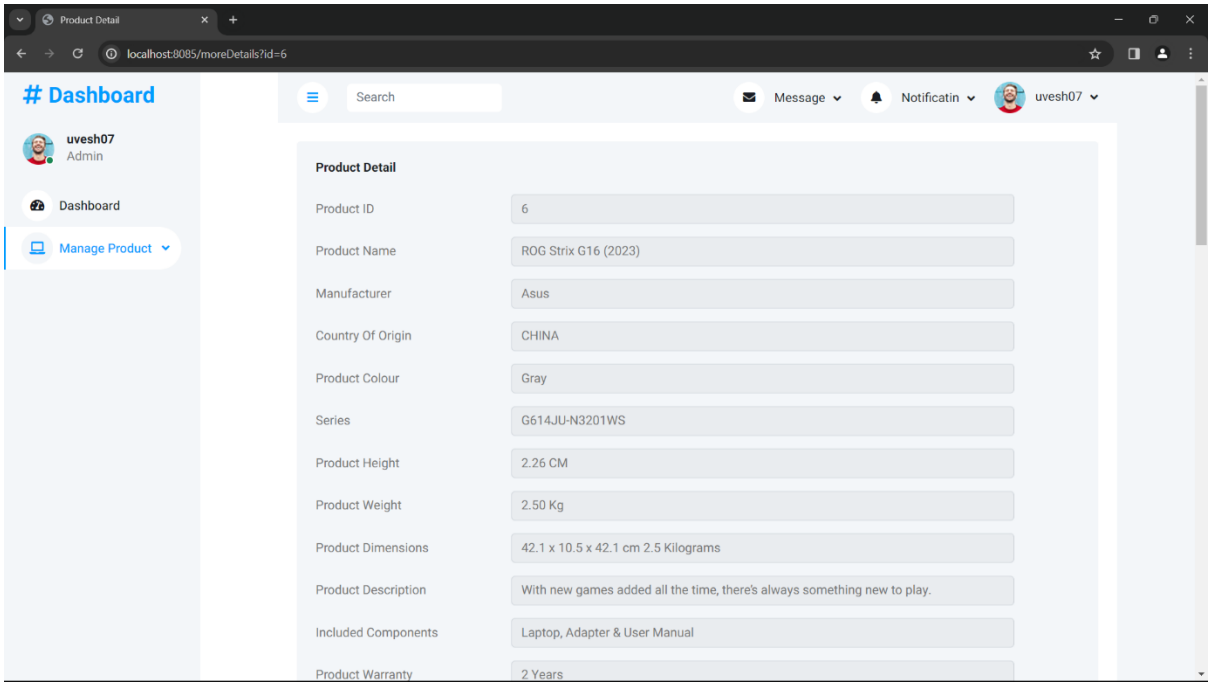


Figure 30 More details page

5. References

1. https://spring.io/guides/gs/serving-web-content
2. https://www.geeksforgeeks.org/spring-mvc/
3. https://www.toptal.com/spring/beginners-guide-to-mvc-with-spring-framework
4. https://www.youtube.com/watch?v=Ku3gsv7_bCc
5. https://www.youtube.com/watch?v=6zfIxgaVkQI
6. https://spring.io/projects/spring-framework
7. https://www.baeldung.com/thymeleaf-in-spring-mvc